PROJECT DOCUMENTATION

DISASTER MANAGEMENT ANALYTICS

Data Analysis, Visualization and Interactive Streamlit Application



Submitted by

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1. Project Overview

Disaster Management Analytics is a data analytics project designed to organize and examine disaster-related information. It brings together incident records, emergency response data, shelter information, relief distribution, population details, resource availability, weather observations and communication coverage.

The project combines Python-based data processing and visualization with an interactive Streamlit interface. A Power BI dashboard is also included to support visual exploration of the prepared data.

2. Problem Statement

During a disaster, decision-makers need a clear view of incident patterns, severity, response performance, shelter pressure, relief gaps and available resources. Considering these datasets separately can make it difficult to identify locations that may require additional attention. This project provides an integrated analytical view.

3. Objectives

- Provide a frontend through which users can upload the required CSV datasets.
- Analyze disaster incidents and their severity distribution.
- Examine emergency response times and response-team performance.
- Compare shelter capacity with occupancy and identify available capacity.
- Review relief distribution and resource readiness.
- Explore locality-level risk indicators and communicate findings through charts and dashboards.

4. Technologies Used

- Python – application logic and data analysis
- Streamlit – interactive web application
- Pandas – data loading, cleaning and transformation
- Matplotlib / Seaborn – charts and visualizations
- Power BI – dashboard-based reporting
- Git and GitHub – version control and project hosting

5. Dataset Description

Dataset	Purpose
incidents.csv	Disaster incident records and severity information
shelters.csv	Shelter capacity and occupancy details
responses.csv	Emergency response records and response-time analysis
relief.csv	Relief item distribution and fulfillment information
population.csv	Population and locality-level context
resources.csv	Resource availability and readiness information
weather.csv	Weather observations associated with the analysis
communication.csv	Communication coverage information

6. Methodology

The project follows a structured workflow to transform the source datasets into analytical outputs:

- **Dataset preparation:** Load the CSV files and inspect their structure and fields.
- **Data processing:** Clean and standardize locality names and prepare fields needed for analysis.
- **Analytical calculations:** Derive incident counts, severity summaries, response-time measures, shelter utilization, relief fulfillment and resource readiness indicators.
- **Risk assessment:** Combine selected locality-level indicators into a weighted risk score.
- **Visualization:** Present results through charts, an interactive Streamlit interface and the Power BI dashboard.

The locality-level Risk_Score used in the analysis combines incident frequency (25%), severity (25%), vulnerability (20%), readiness gap (15%) and shelter pressure (15%). The score is an analytical prioritization indicator and should be interpreted alongside the underlying data and local context.

7. Main Analytical Areas

- Incidents by locality and severity.
- Response-time distribution and response-team comparison.
- Shelter capacity versus occupancy.
- Relief gaps by item.
- Locality-level risk comparison.
- Resource readiness and communication coverage.

8. Streamlit Application – How to Run

Clone the repository and install the dependencies:

```
git clone https://github.com/sayanbiswas425/Disaster-Management-Analytics.git

cd Disaster-Management-Analytics

python -m pip install -r requirements.txt

python -m streamlit run app.py
```

After the application starts, open the local URL shown in the terminal. Upload the required CSV datasets through the frontend and run the available analysis.

9. Project Repository

https://github.com/sayanbiswas425/Disaster-Management-Analytics

10. Conclusion

The project demonstrates how multiple disaster-related datasets can be brought together for analysis and visualization. The Streamlit frontend provides an interactive way to upload data and explore analytical outputs, while the Power BI dashboard offers an additional reporting view. The resulting insights can support discussion and planning; operational decisions should always be validated by responsible authorities using current field information.

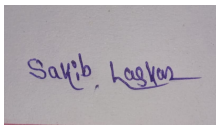
11. Team Members and Signatures

The following team members contributed to the project:

SAYAN BISWAS
Team Member



SAKIB LASKAR
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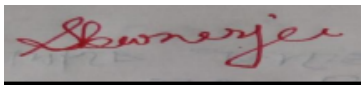


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